

NEW-AGE GLOBAL UNIVERSITY FOR LIBERAL EDUCATION



Orthographic Projection of Lines



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PROJECTIONS OF STRAIGHT LINES.

INFORMATION REGARDING A LINE MEANS IT'S LENGTH, POSITION OF IT'S ENDS WITH HP & VP, IT'S INCLINATIONS WITH HP & VP WILL BE GIVEN.

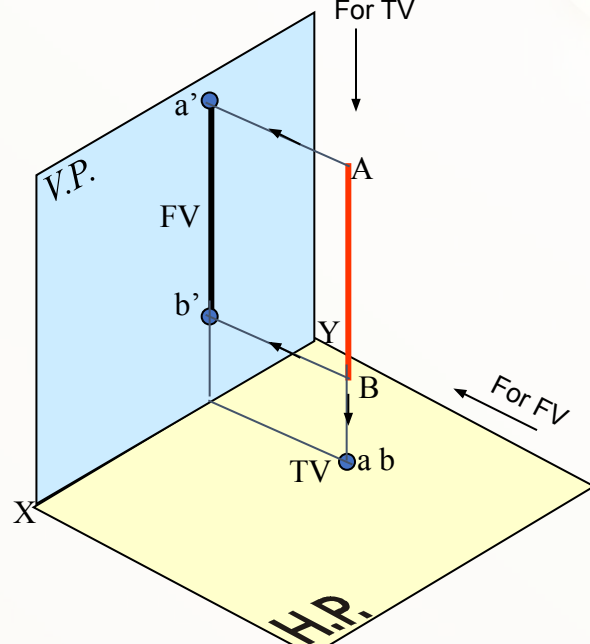
AIM:- TO DRAW IT'S PROJECTIONS - MEANS FV & TV.

SIMPLE CASES OF THE LINE

- 1. A VERTICAL LINE (LINE PERPENDICULAR TO HP & // TO VP)**
- 2. LINE PARALLEL TO BOTH HP & VP.**
- 3. LINE INCLINED TO HP & PARALLEL TO VP.**
- 4. LINE INCLINED TO VP & PARALLEL TO HP.**
- 5. LINE INCLINED TO BOTH HP & VP.**

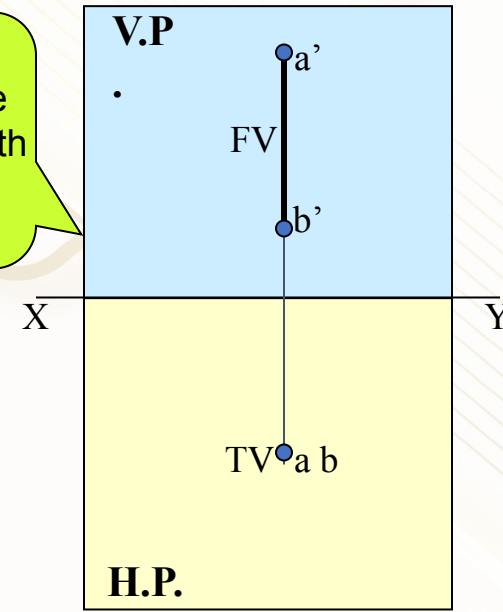
1

A Line
perpendicular
to HP
&
// to VP



Note:

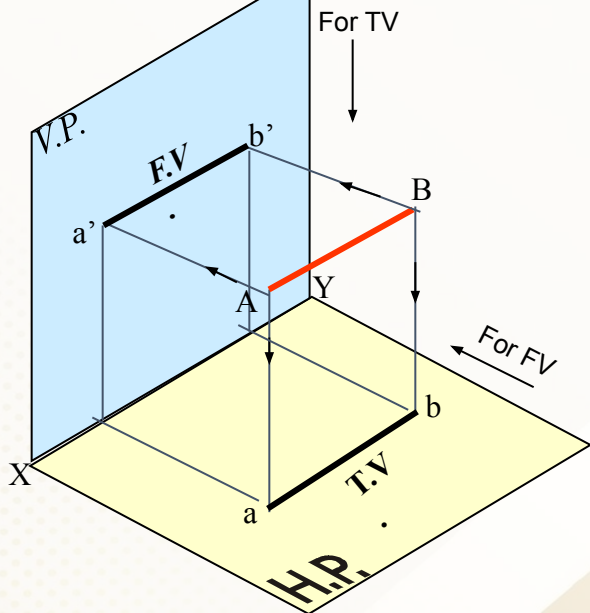
FV is a vertical line
Showing True Length
&
TV is a point.



Orthographic Pattern

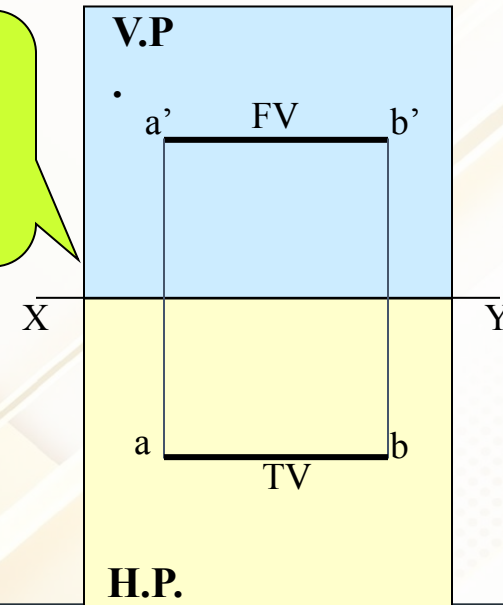
2

A Line
// to HP
&
// to VP



Note:

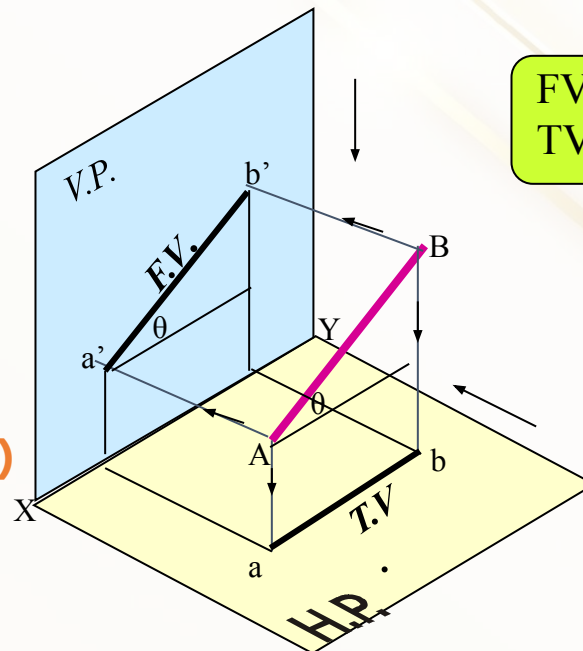
FV & TV both are
// to xy
&
both show T. L.



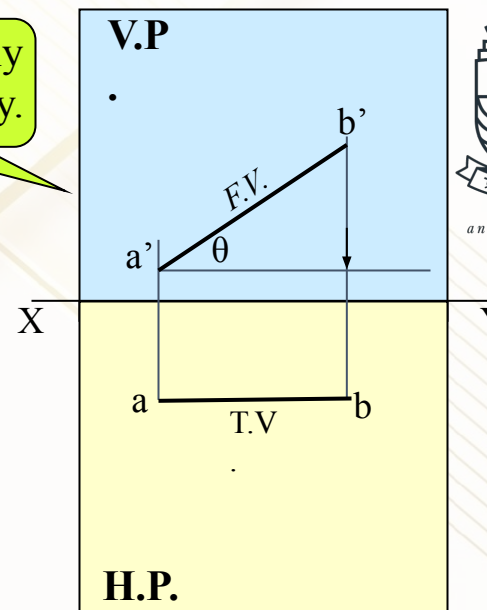
3

A Line inclined to HP
and
parallel to VP

(Pictorial presentation)



FV inclined to xy
TV parallel to xy.

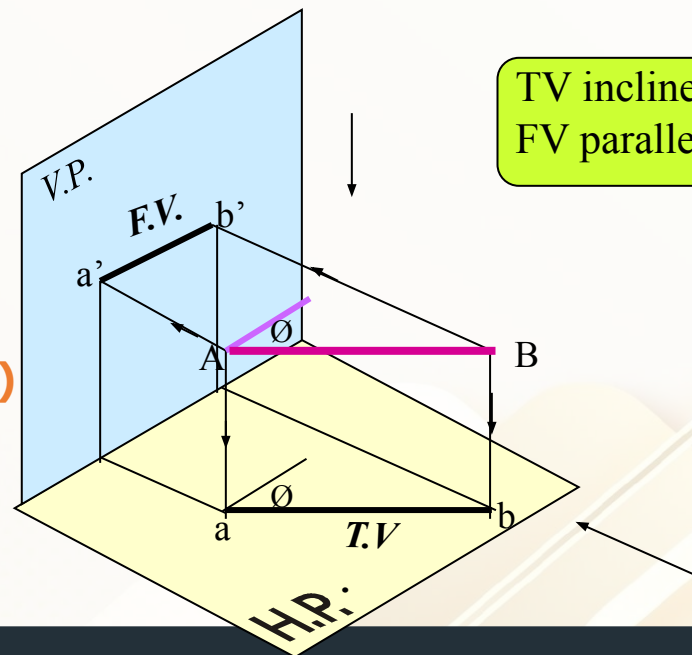


Orthographic Projections

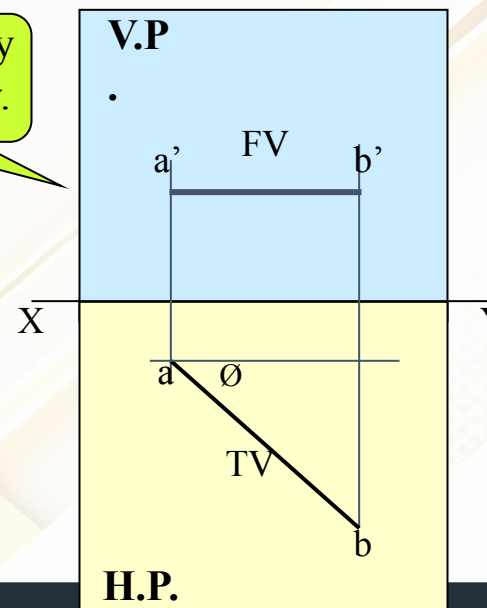
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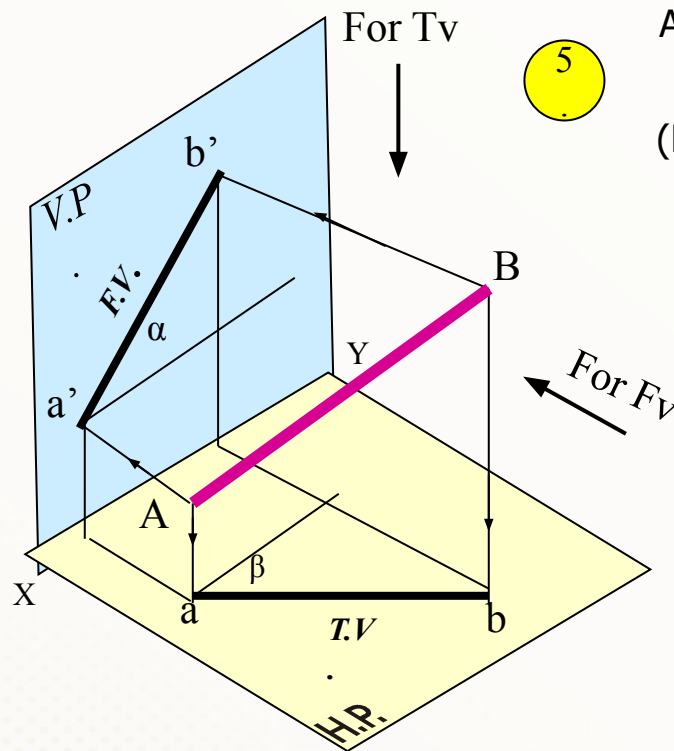
A Line inclined to VP
and
parallel to HP

(Pictorial presentation)



TV inclined to xy
FV parallel to xy.



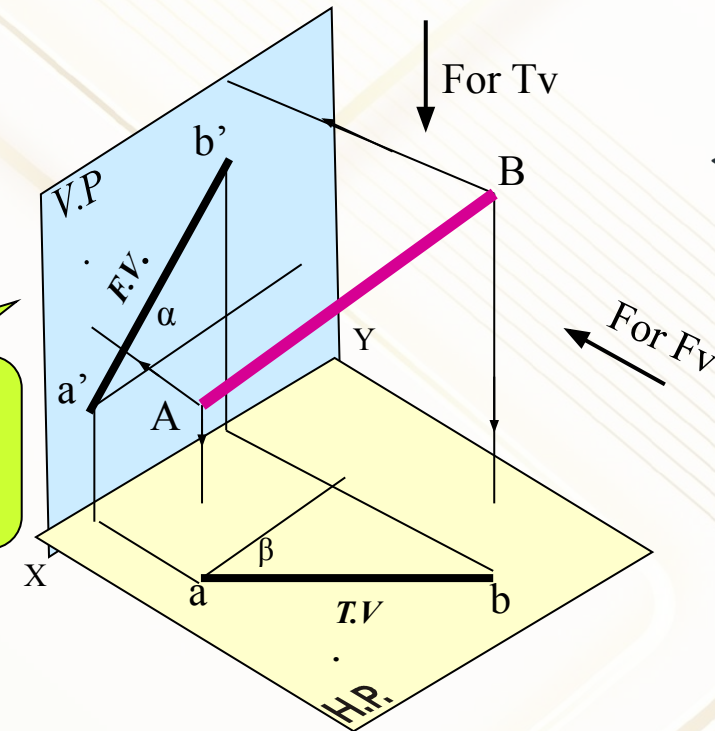


5

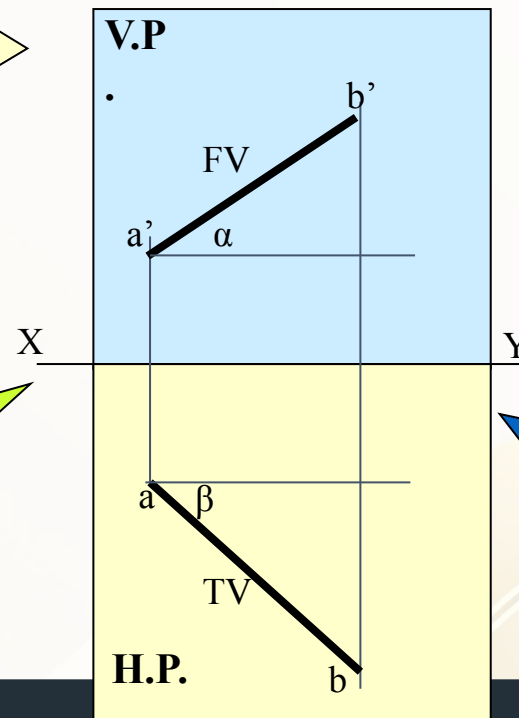
A Line inclined to both
Hp and Vp
(Pictorial presentation)

On removal of object
i.e. Line AB

Fv as a image on Vp.
Tv as a image on Hp,

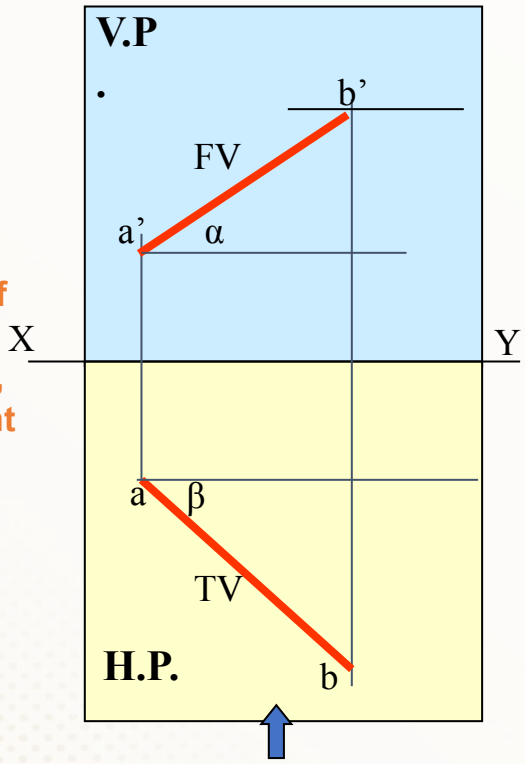


Orthographic Projections
Fv is seen on Vp clearly.
To see Tv clearly, HP is
rotated 90° downwards,
Hence it comes below xy.



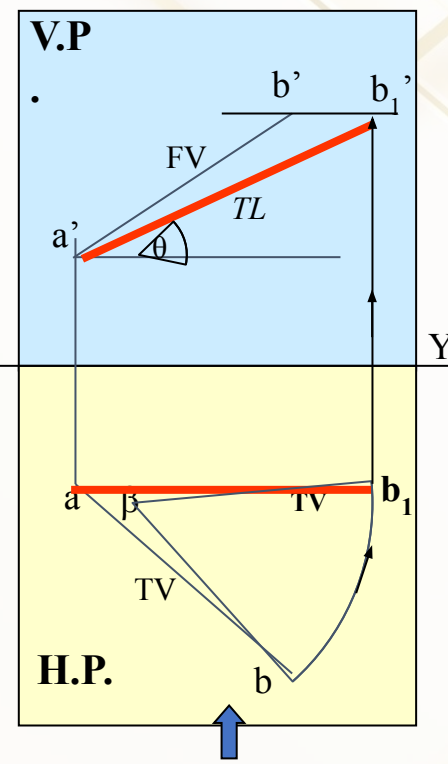
Note These Facts:-
Both FV & TV are inclined to xy.
(No view is parallel to xy)
Both FV & TV are reduced lengths.
(No view shows True Length)

Orthographic Projections
Means FV & TV of Line AB are shown below, with their apparent Inclinations α & β



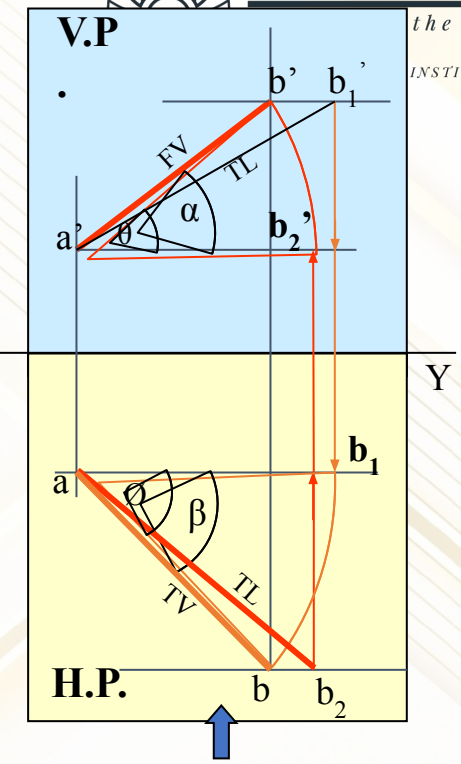
Here TV (ab) is not // to XY line
Hence it's corresponding FV $a'b'$ is **not** showing **True Length & True Inclination with HP.**

Note the procedure
When **FV & TV** known,
How to find True Length.
(Views are rotated to determine True Length & it's inclinations with HP & VP).



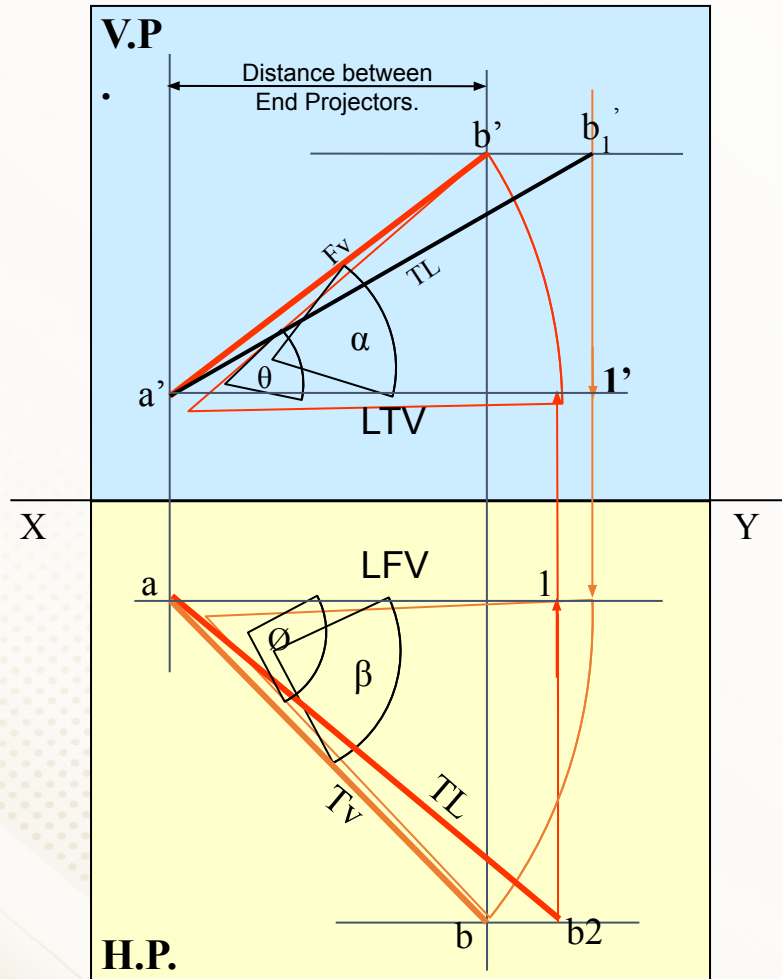
In this sketch, TV is rotated and made // to XY line.
Hence it's corresponding FV $a'b_1'$ is showing **True Length & True Inclination with HP.**

Note the procedure
When True Length is known,
How to locate FV & TV.
(Component $a'b_1'$ of TL is drawn which is further rotated to determine FV)



Here $a'b_1'$ is component of TL ab_1' gives length of **FV**.
Hence it is brought Up to Locus of a' and further rotated to get point b' . $a'b'$ will be FV.
Similarly drawing component of other TL($a'b_1'$) TV can be drawn.

The most important diagram showing graphical relations among all important parameters of this topic. Study and memorize it as a **circuit diagram** and use in solving various problems.



- 1) True Length (TL) – $a'b_1'$ & ab_1
- 2) Angle of TL with Hp – θ
- 3) Angle of TL with Vp – ϕ
- 4) Angle of FV with xy – α
- 5) Angle of TV with xy – β
- 6) LFV (length of FV) – Component ($a-1$)
- 7) LTV (length of TV) – Component ($a'-1'$)
- 8) Position of A- Distances of a & a' from xy
- 9) Position of B- Distances of b & b' from xy
- 10) Distance between End Projectors

**Important
TEN parameters
to be remembered
with Notations
used here onward**

NOTE this

θ & α Construct with a'
 ϕ & β Construct with a
 b' & b_1' on same locus.
 b & b_1 on same locus.

Also Remember

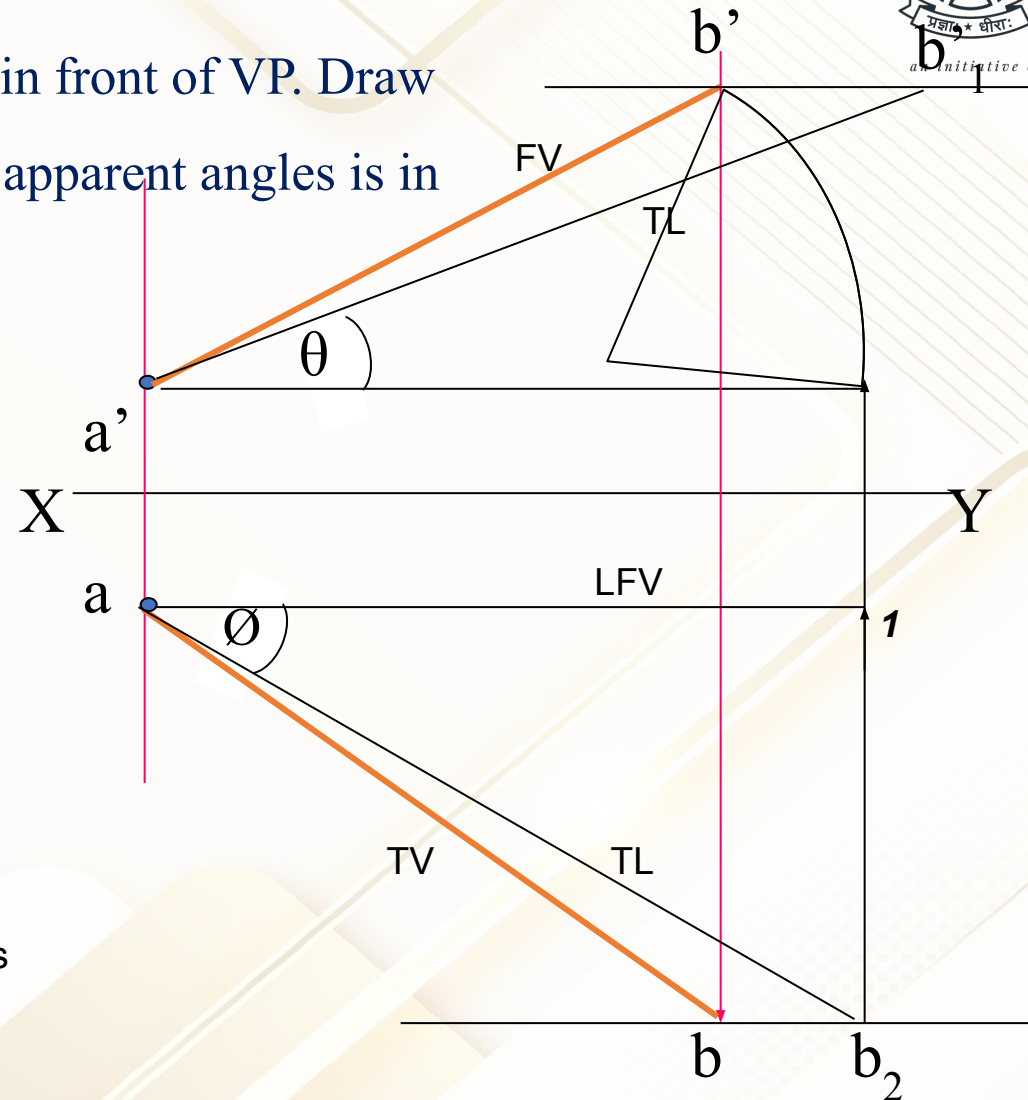
True Length is never rotated. It's horizontal component is drawn & it is further rotated to locate view.

Views are always rotated, made horizontal & further extended to locate TL, θ & ϕ

Problem 1. Line AB is 75 mm long and it is 30° & 40° Inclined to HP & VP respectively. End A is 20 mm above HP and 30 mm in front of VP. Draw projections of the Line and find apparent lengths and apparent angles in 1st quadrant.

SOLUTION STEPS:

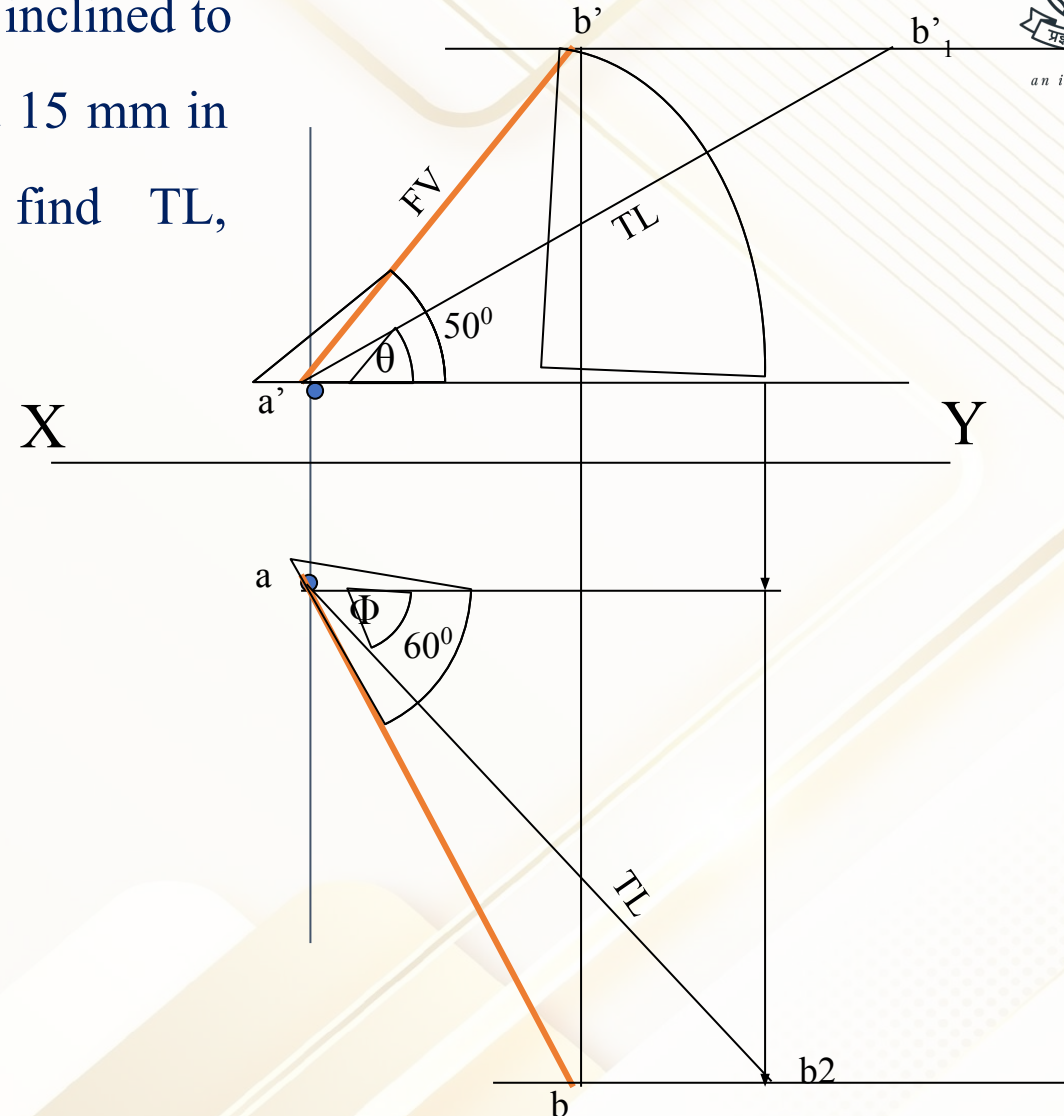
- 1) Draw xy line and one projector.
- 2) Locate a' 12mm above xy line & a 10mm below xy line.
- 3) Take 30° angle from a' & 40° from a and mark TL i.e. 75mm on both lines. Name those points b_1' and b_1 respectively.
- 4) Join both points with a' and a resp.
- 5) Draw horizontal lines (Locus) from both points.
- 6) Draw horizontal component of TL a b_1 from point b_1 and name it 1.
(the length $a-1$ gives length of Fv as we have seen already.)
- 7) Extend it up to locus of a' and rotating a' as center locate b' as shown. Join $a' b'$ as Fv.
- 8) From b' drop a projector down ward & get point b . Join a & b i.e. Tv.



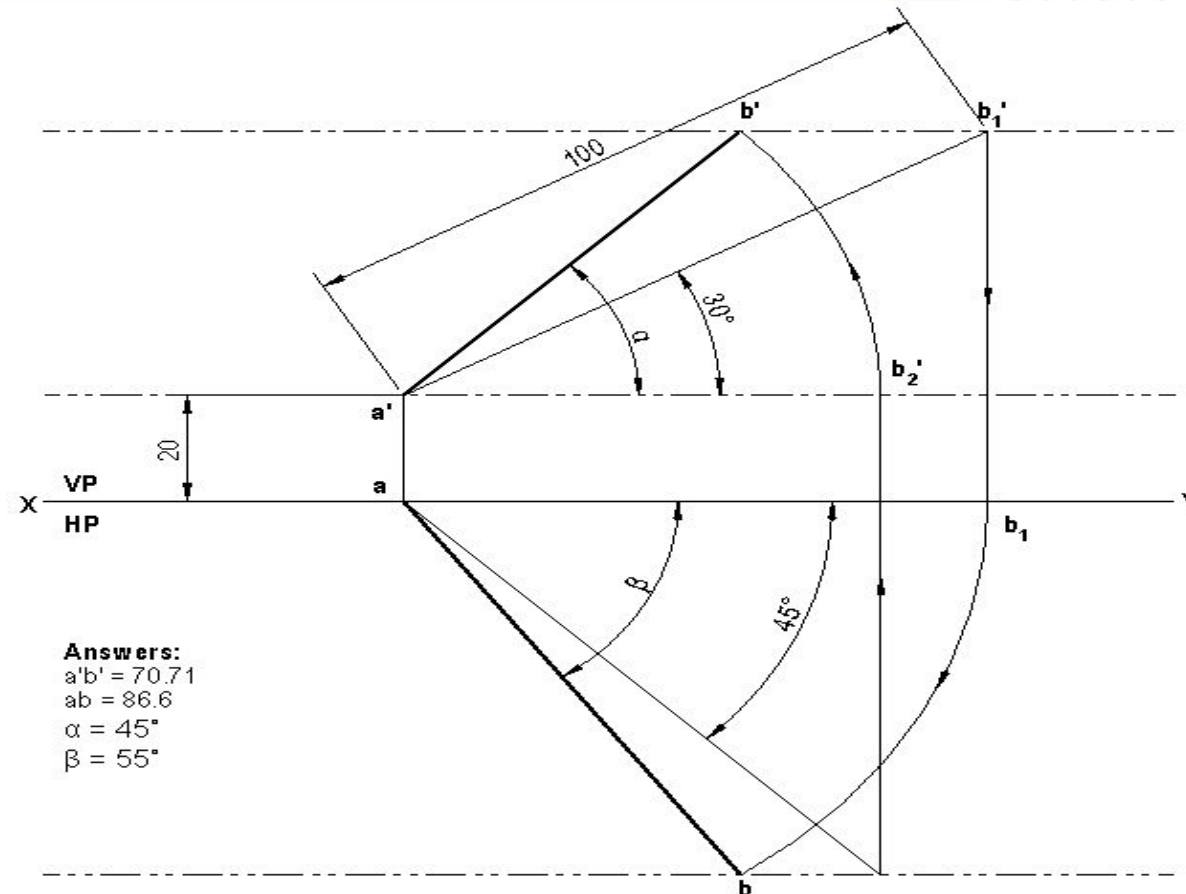
Problem 2. FV of line AB is 50° inclined to XY and measures 55 mm long while it's TV is 60° inclined to XY line. If end A is 10 mm above HP and 15 mm in front of VP, draw it's projections, find TL, inclinations of line with HP & VP.

SOLUTION STEPS:

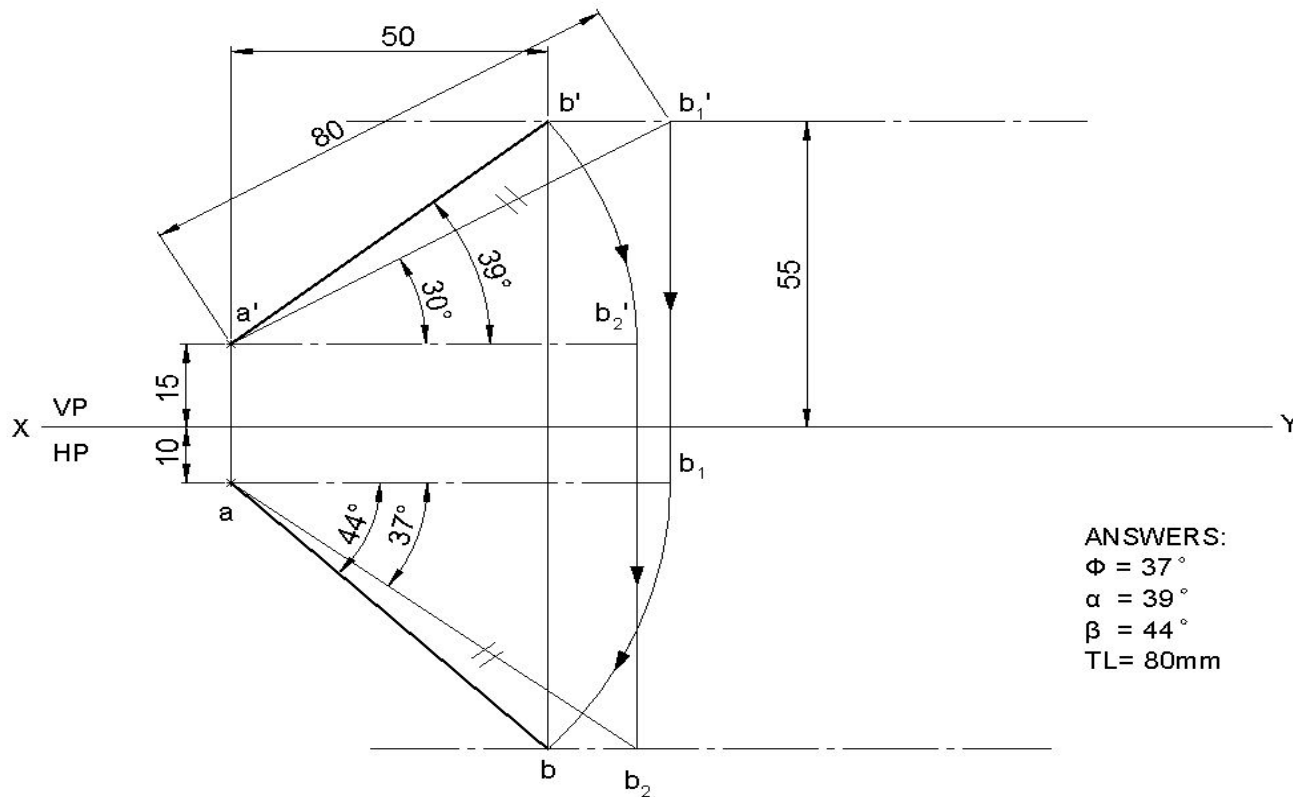
1. Draw xy line and one projector.
2. Locate a' 10 mm above xy and a 15 mm below xy line.
3. Draw locus from these points.
4. Draw FV 50° to xy from a' and mark b' Cutting 55 mm on it.
5. Similarly draw TV 60° to xy from a & drawing projector from b' Locate point b and join a b.
6. Then rotating views as shown, locate True Lengths ab_1 & $a'b_1'$ and their angles with HP and VP.



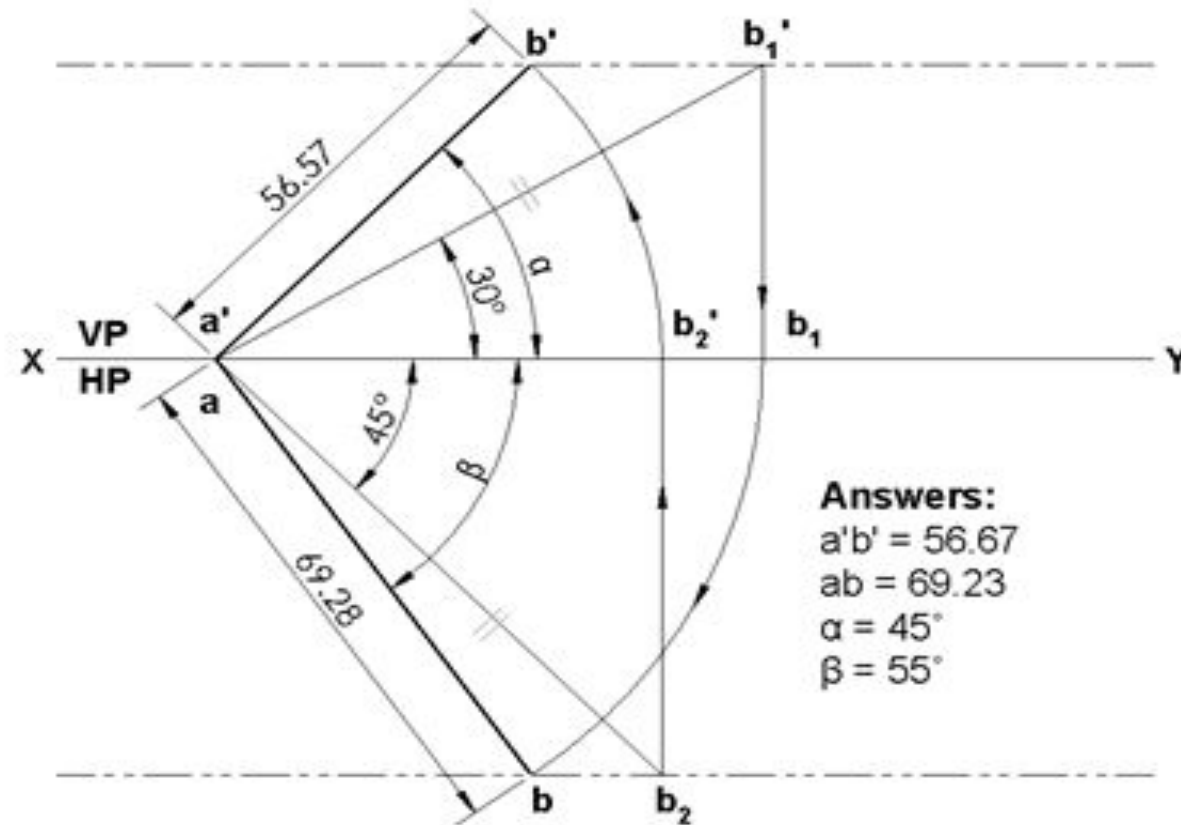
Problem 3. Draw the projections of a line while AB 100 mm long inclined at 45° to VP and 30° to HP. One end of the line is 20 mm above the HP and in the VP. Also determine the apparent length and inclinations.



Problem 4. A line has its end A 15 mm above HP and 10 mm in front of VP. The end B is 55 mm above HP and the line is inclined at 30° to HP. The distance between the end projector is 50 mm. Draw the projections of the line and determine the true length of the line and its inclination with VP.



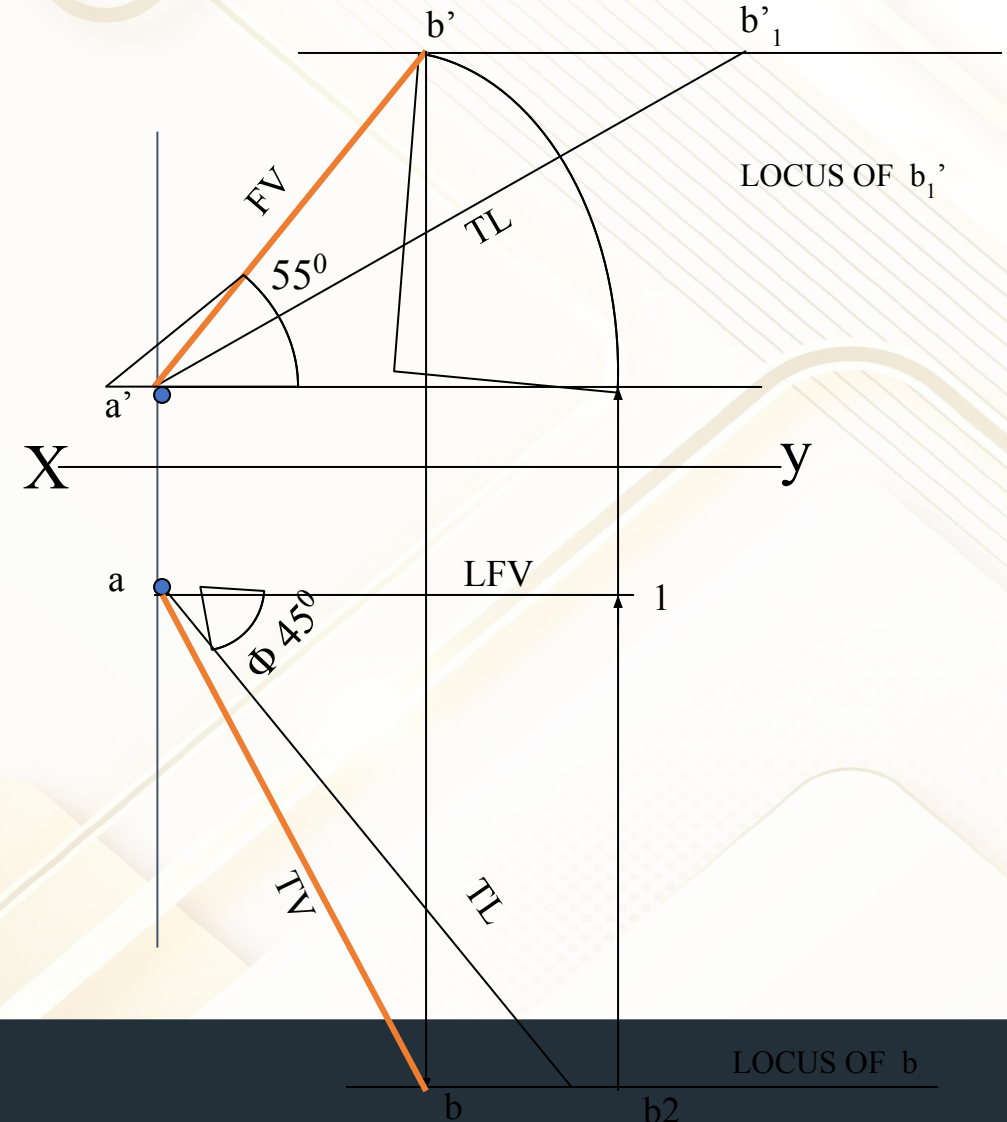
Problem 5. A line AB 80 mm long is inclined to HP at 30° and inclined to VP at 45° . Draw front and top views of line and determine their lengths. Also measure the perpendicular distance of end B from both HP and VP.



Problem 6. Line AB 75mm long makes 45° inclination with VP while it's FV makes 55° . End A is 10 mm above HP and 15 mm in front of VP. If line is in 1st quadrant draw it's projections and find it's inclination with HP.

Solution Steps:-

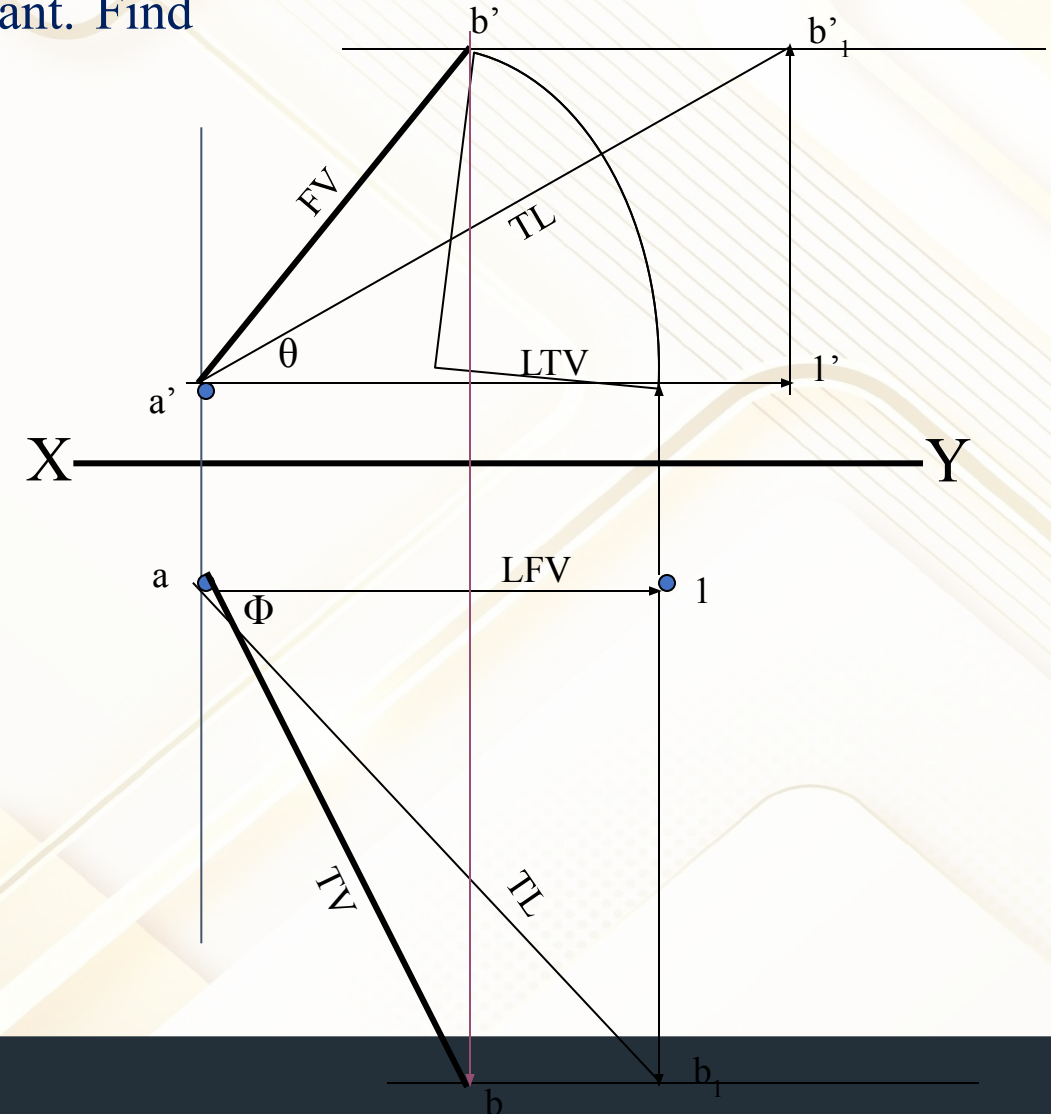
1. Draw x-y line.
2. Draw one projector for a' & a
3. Locate a' 10 mm above x-y & TV a 15 mm below xy.
4. Draw a line 45° inclined to xy from point a and cut TL 75 mm on it and name that point b_1 . Draw locus from point b_1
5. Take 55° angle from a' for FV above xy line.
6. Draw a vertical line from b_1 up to locus of a and name it 1. It is horizontal component of TL & is LFV.
7. Continue it to locus of a' and rotate upward up to the line of FV and name it b' . This $a'b'$ line is FV.
8. Drop a projector from b' on locus from point b_1 and name intersecting point b . Line ab is Tv of line AB.
9. Draw locus from b' and from a' with TL distance cut point b_1'
10. Join $a'b_1'$ as TL and measure it's angle at a' . It will be true angle of line with HP.



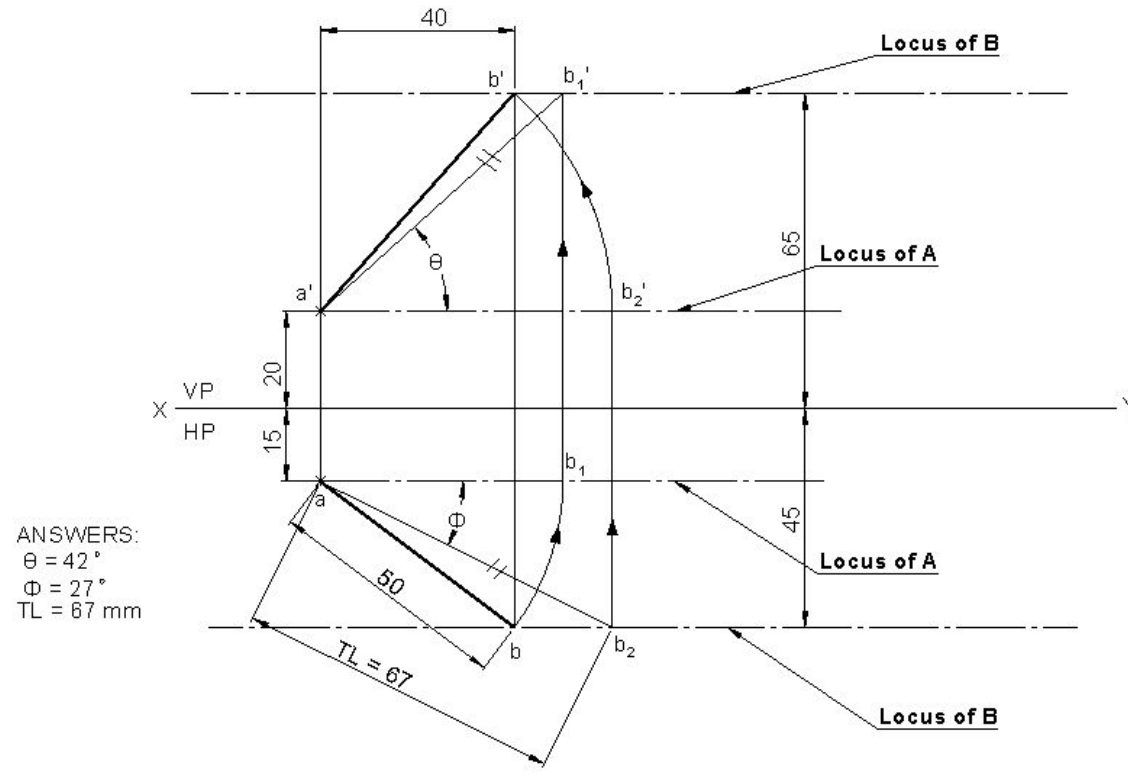
Problem 7. Line AB is 75 mm long .It's FV and TV measure 50 mm & 60 mm long respectively. End A is 10 mm above HP and 15 mm in front of VP. Draw projections of line AB if end B is in first quadrant. Find angle with HP and VP.

SOLUTION STEPS:

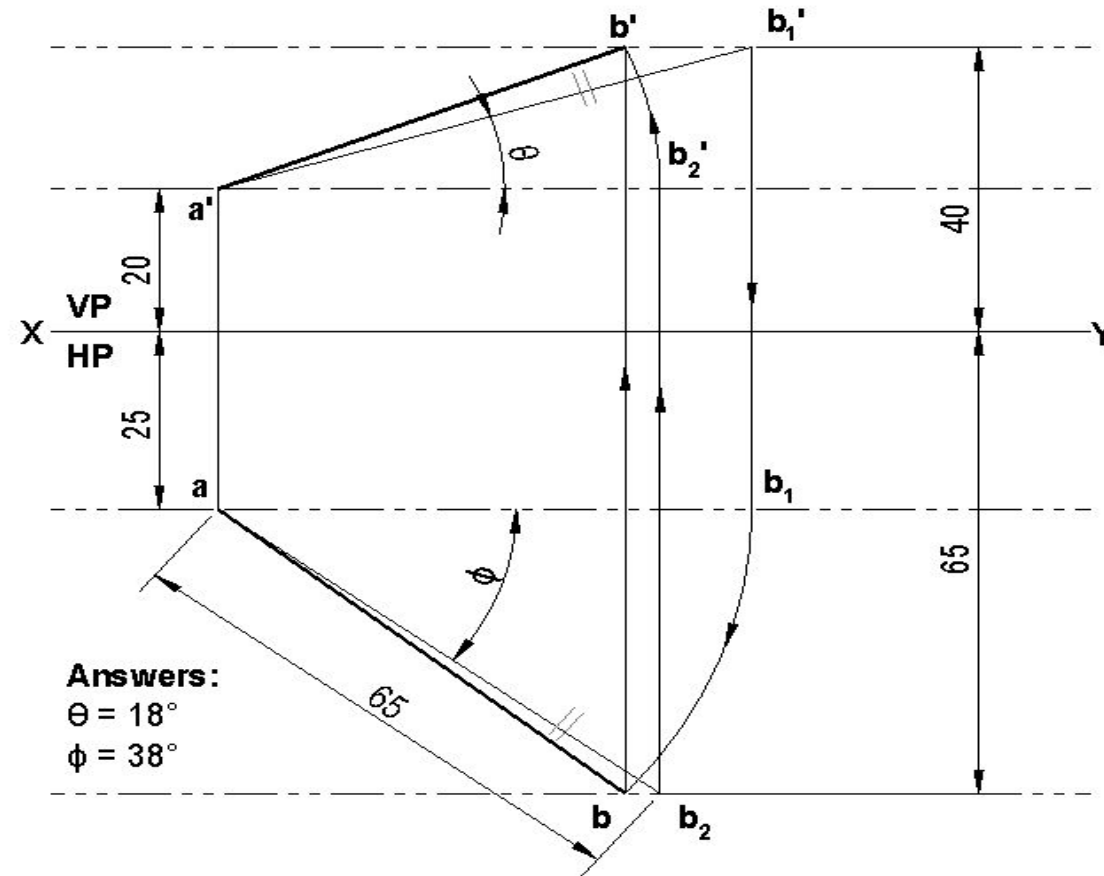
1. Draw xy line and one projector.
2. Locate a' 10 mm above xy and a 15 mm below xy line.
3. Draw locus from these points.
4. Cut 60mm distance on locus of a' & mark $1'$ on it as it is LTV.
5. Similarly cut 50mm on locus of a and mark point 1 as it is LFV.
6. From $1'$ draw a vertical line upward and from a' taking TL (75mm) in compass, mark b'_1 point on it. Join $a' b'_1$ points.
7. Draw locus from b'_1
8. With same steps below get b_1 point and draw also locus from it.
9. Now rotating one of the components i.e. a-1 locate b' and join a' with it to get FV.
10. Locate TV similarly and measure Angles θ & Φ .



Problem 8. The distance between the end projectors through the end points of a line AB is 40 mm. The end A is 20 mm above HP and 15 mm in front of VP. The end B is 45 mm in front of VP and 65 mm above HP. Complete the projections. Find the true length of the line and its inclination with HP and VP.

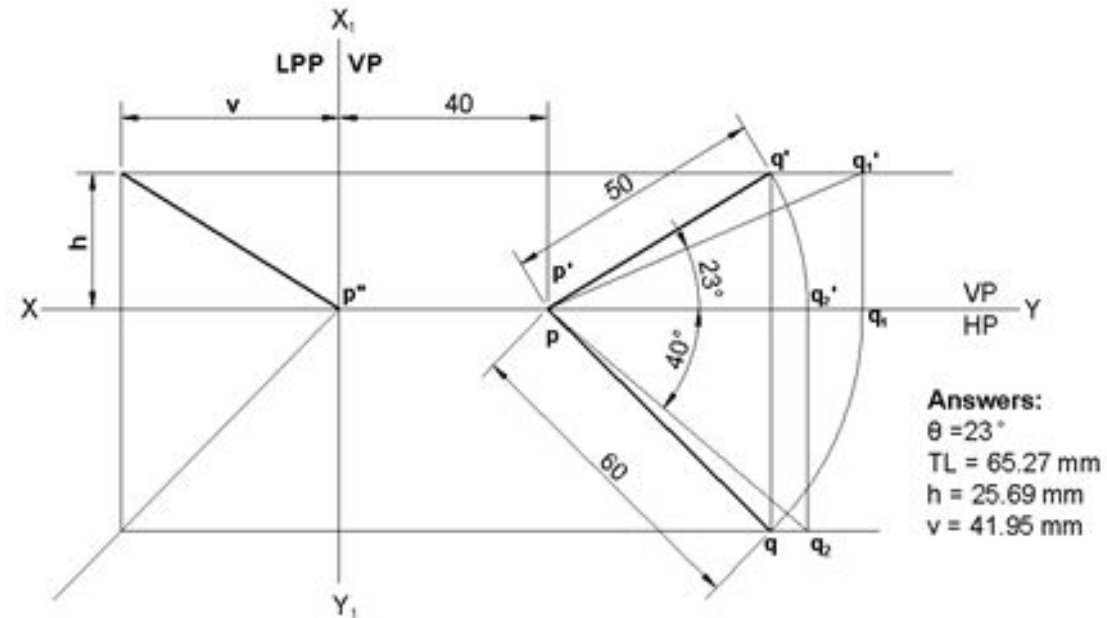


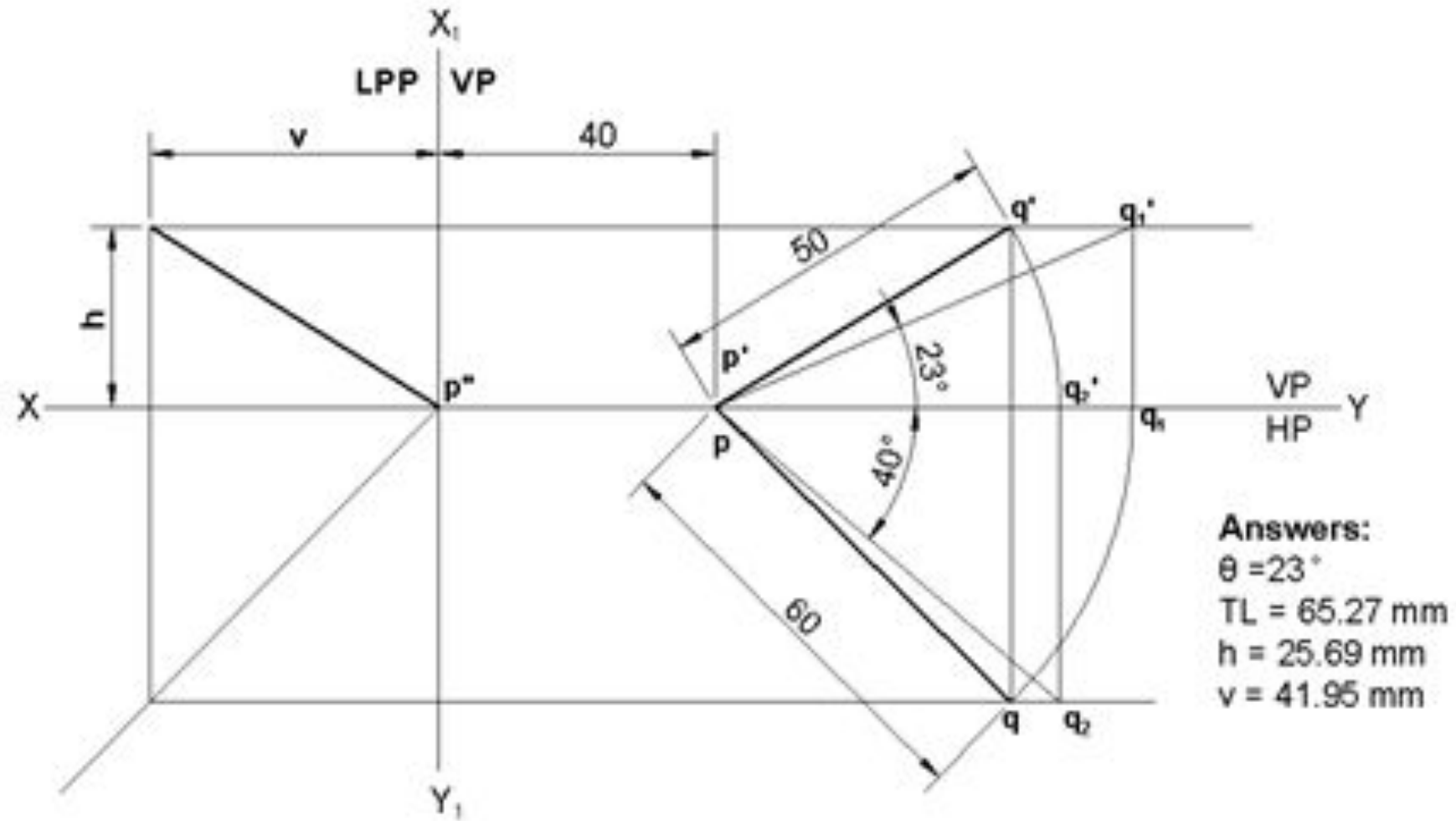
Problem 9. A line AB, 65 mm, has its end A 20 mm above HP and 25 mm in front of the VP. The end B is 40 mm above the HP and 65 mm in front of VP. Draw the projections of AB and show its inclinations with HP and VP.



Problem 10. A straight-line PQ inclined at 40° to VP has $p\ q = 60$ mm and $p'\ q' = 50$ mm. The end P is both in HP and VP, and 40 mm to the right of left profile plane

- Draw the projections of the straight-line PQ
- Find the true length and true inclination with HP
- Draw the profile view of the straight line
- Find the position of the end Q with HP and VP





Thank you